

UNIT - III

SCIENTIFIC VALUES

Q1. Explain the method and process of hypothesis testing. How are facts validated using an evidence-based approach?

Scientific thinking is a process of reviewing ideas using **science, observation, investigation and testing** to gain knowledge.

A scientist does not simply accept a claim as true. The claim is checked using **evidence, experiments, observations and data**.

What is a Hypothesis?

A **hypothesis** is a possible explanation or prediction about something that can be tested.

It is formed using:

- Observations
- Existing knowledge
- Previous evidence

A good hypothesis should be possible to **test and verify or reject** through empirical testing.

Example:

Suppose a student says: "**Using a particular study method improves students' marks.**"

This is a hypothesis because we can test it by collecting data from students.

Method and Process of Hypothesis Testing

Step 1: Observation and Questioning

Scientific investigation begins with **careful observation** of a phenomenon or collection of data.

Based on the observation, a question is formed.

Example: "Does regular practice improve students' marks?"

Step 2: Formulation of Hypothesis

Based on observation and existing knowledge, a hypothesis is formed.

Example: "Students who practise regularly will get higher marks."

A hypothesis should be possible to test using empirical evidence.

Step 3: Designing the Experiment

A suitable experiment or study is designed to test the hypothesis.

The researcher selects suitable participants, collects relevant data and tries to reduce bias.

Step 4: Data Collection

Relevant data is collected through:

- Observation
- Experiments
- Measurements
- Surveys

The collected data is used to test the hypothesis.

Step 5: Data Analysis

The collected data is analysed using **statistical or other analytical methods**.

The researcher looks for patterns, trends and relationships in the data.

Step 6: Interpretation and Conclusion

The results are compared with the prediction made by the hypothesis. If the evidence supports the hypothesis, it receives support.

If the evidence does not support it, the hypothesis may be rejected or modified.

Validation of Facts Using Evidence-Based Approach

Evidence-based validation means checking whether a claim is supported by **reliable evidence and data**, instead of accepting it based only on personal belief.

The process includes:

- 1. Identify the Claim:** First, clearly identify the fact or claim that needs to be tested.
 - 2. Research Design:** Design an experiment or study to test the claim.
 - 3. Data Collection:** Collect relevant and reliable data.
 - 4. Data Analysis:** Analyse the collected data to find relationships and significant results.
 - 5. Review and Replication:** The findings are evaluated through **peer review**, and other researchers can repeat the experiment to confirm the results.
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Why Evidence-Based Validation is Important

Evidence-based validation is important because it:

- 1. Reduces personal bias**
- 2. Provides objectivity**
- 3. Makes facts more credible and reliable**
- 4. Helps build scientific knowledge step by step**

Thus, these methods help develop accurate and reliable scientific knowledge.

Q2. Explain Inductive and Deductive Thinking with suitable examples.

Inductive and deductive thinking are two important methods of scientific reasoning.

They help us to:

- Understand observations
- Identify patterns
- Form ideas
- Test ideas
- Reach logical conclusions

The main difference is the direction of reasoning:

Inductive → Specific to General

Deductive → General to Specific

Inductive Thinking

Inductive reasoning is a logical process based on **experiences, observations and facts**.

It starts with specific observations and uses them to form a **general conclusion or theory**.

Flow: Specific Observations → Pattern → General Conclusion

Example for Inductive Thinking

Suppose a person observes that several low-cost editing apps experience glitches.

- App A → Glitches
- App B → Glitches
- App C → Glitches

After observing this pattern, the person may conclude:

"Low-cost editing apps may experience glitches."

This is **inductive thinking** because specific observations are used to form a general idea.

Steps of Inductive Thinking

The notes explain three main stages:

- 1. Observation:** Observe a phenomenon or collect information.
- 2. Observe a Pattern:** Identify a repeated pattern in the observations.
- 3. Develop a Theory:** Use the observed pattern to form a general theory or conclusion.

Deductive Thinking

Deductive reasoning, also called **top-down reasoning**, starts with general statements or premises and reaches a specific conclusion.

Flow: General Rule → Specific Case → Conclusion

Example for Deductive Thinking

Consider:

Premise 1: All men have ten fingers.

Premise 2: John is a man.

Conclusion: Therefore, John has ten fingers. Here, we start with a general statement and apply it to a specific person. Therefore, it is **deductive thinking**.

Difference Between Inductive and Deductive Thinking

Inductive Thinking	Deductive Thinking
Specific → General	General → Specific
Starts with observations	Starts with general premises
Finds patterns	Applies a general rule
Develops theories	Tests or applies theories
Conclusion is probable	Conclusion follows logically when premises are valid
Useful for forming hypotheses	Useful for testing hypotheses

Strengths and Limitations

Inductive Thinking

Strengths:

- Helps generate hypotheses.
- Identifies patterns and trends.
- Helps make predictions.

Limitation:

- The conclusion is not absolutely certain.
- A general conclusion may not be true in every situation.

Deductive Thinking

Strengths:

- Provides logical conclusions.
- Useful for testing hypotheses.
- Applies general principles to specific situations.

Limitation:

- If the starting premise is wrong, the conclusion can also be wrong.